

Interactive Solar System for Kids

Game Programming | Short in class assignment handout



In this class activity, I want you to turn your solar system scene into a small interactive experience for kids. A child should be able to click a planet or a moon and get a clear, fun response.

Time

75 minutes

Marks

20 total

Focus

Clickable planets and moons

Task

- Start from the solar system scene we built in class.
- Include the Sun, at least one planet, and at least one moon.
- Make at least one planet and one moon clickable.
- When the user clicks, add a meaningful response that helps the child explore the scene.

Keep the scope realistic. A simple polished idea is better than a large unfinished idea.

Minimum requirements

- Use rotation or orbit motion.
- Change the camera when an object is selected.
- Add one visual response or one audio response.
- Give the player one way to return to the main view.
- Use at least four class ideas, such as materials, lighting, behaviors, audio, cameras, or prefabs.

Good ideas

- Zoom the camera toward the selected object.
- Make the camera look at the selected object.
- Show a short fact in simple child friendly language.
- Play a sound effect when the object is clicked.
- Change glow, color, or scale for a moment.
- Add a bonus such as a comet or a mini map.

What to submit

- Show your scene in Play mode.
- Submit the scene and any scripts you created.
- Be ready to explain your idea in about one minute.

Rubric and Helpful Links

Use this page while you build and test your scene



Quick rubric

Total: 20 marks

Criterion	4 marks	2 to 3 marks	0 to 1 mark
Interaction	A planet and a moon respond clearly to clicks. The interaction feels intentional and works well.	The main interaction works, but one part is basic or not fully polished.	The click response is unclear, missing, or does not work reliably.
Camera use	The camera change helps the player explore the scene and there is a clear return view.	The camera changes, but the movement or return view is limited.	Camera use is weak, confusing, or missing.
Use of class concepts	The project uses at least four class ideas well, such as materials, lighting, behaviors, audio, cameras, or prefabs.	The project uses some class ideas, but the connection is only partly developed.	There is little evidence of the class topics in the final scene.
Kid friendly design	The experience is playful, simple, and suitable for a child audience.	The idea is interesting, but the child friendly focus could be stronger.	The design is unclear or not suitable for the target audience.
Technical quality and completion	The scene is organized, named clearly, and ready to demo in Play mode.	The scene mostly works, but there are a few technical or presentation issues.	The work is incomplete, unstable, or difficult to demonstrate.

Note: A small polished prototype can score higher than a large unfinished scene.

Helpful links

If you get stuck, these official pages are useful. You may use **OnMouseDown()** or a camera raycast to detect clicks.

Resource	Link
Cameras overview	https://docs.unity3d.com/Manual/CamerasOverview.html
Click with OnMouseDown()	https://docs.unity3d.com/ScriptReference/MonoBehaviour.OnMouseDown.html
Click with Physics.Raycast	https://docs.unity3d.com/ScriptReference/Physics.Raycast.html
Prefabs	https://docs.unity3d.com/Manual/Prefabs.html
Audio Source	https://docs.unity3d.com/Manual/class-AudioSource.html
Trail Renderer	https://docs.unity3d.com/Manual/class-TrailRenderer.html
NASA Solar System	https://science.nasa.gov/solar-system/

Tip: Copy the links directly from the PDF. They are shown as plain text for easy copy and paste.